TESTING THE BASE CHANGE MAP AFTER PUSHING FORWARD ALONG THE BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The possibility of testing an isomorphism after applying the pushforward along the base changed ring map. Invoked at the start of the counted proof (source lines 2709-2710). Reduces the statement to a comparison after pushing forward.
- The base change map for the derived internal hom and its identification with the map obtained after pushing forward. Invoked inside the counted proof (source lines 2711-2717). Identifies the pushed forward map with the known base change map for the internal hom.
- The unwinding of the two sides of that map along the adjunction between tensor and internal hom. Invoked at the end of the counted proof (source lines 2718-2722). Produces the explicit comparison whose isomorphism is the assertion.

Source reference. Original equation equation-base-change: sources/stacks-dualizing/pinned-original-source.tex, source lines 2669–2669. Navigation pointer only: the counted text refers to the Stacks item equation-base-change, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-equation-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2717–2717. Navigation pointer only: the counted text refers to the Stacks item more-algebra-equation-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's

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Source reference. Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-internal-hom-diagonal-better: sources/stacks-dualizing/pinned-original-source.tex, source lines 2705–2705. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-internal-hom-diagonal-better, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-upgrade-adjoint-tensor-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2721–2721. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-upgrade-adjoint-tensor-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *In the situation above, the map (equation-base-change) is an isomorphism if and only if the map*

$$R\operatorname{Hom}_R(A, K) \otimes^{\mathbf{L}}_R R' \longrightarrow R\operatorname{Hom}_R(A, K \otimes^{\mathbf{L}}_R R')$$

of More on Algebra, Lemma more-algebra-lemma-internal-hom-diagonal-better is an isomorphism.

Proof. To see that the map is an isomorphism, it suffices to prove it is an isomorphism after applying φ'_*. Applying the functor φ'_* to (equation-base-change) and using that $A' = A \otimes^{\mathbf{L}}_R R'$ we obtain the base change map $R\operatorname{Hom}_R(A, K) \otimes^{\mathbf{L}}_R R' \to R\operatorname{Hom}_{R'}(A \otimes^{\mathbf{L}}_R R', K \otimes^{\mathbf{L}}_R R')$ for derived hom of More on Algebra, Equation (more-algebra-equation-base-change-RHom). Unwinding the left and right hand side exactly as in the proof of More on Algebra, Lemma more-algebra-lemma-base-change-RHom and in particular using More on Algebra, Lemma more-algebra-lemma-upgrade-adjoint-tensor-RHom gives the desired result. □